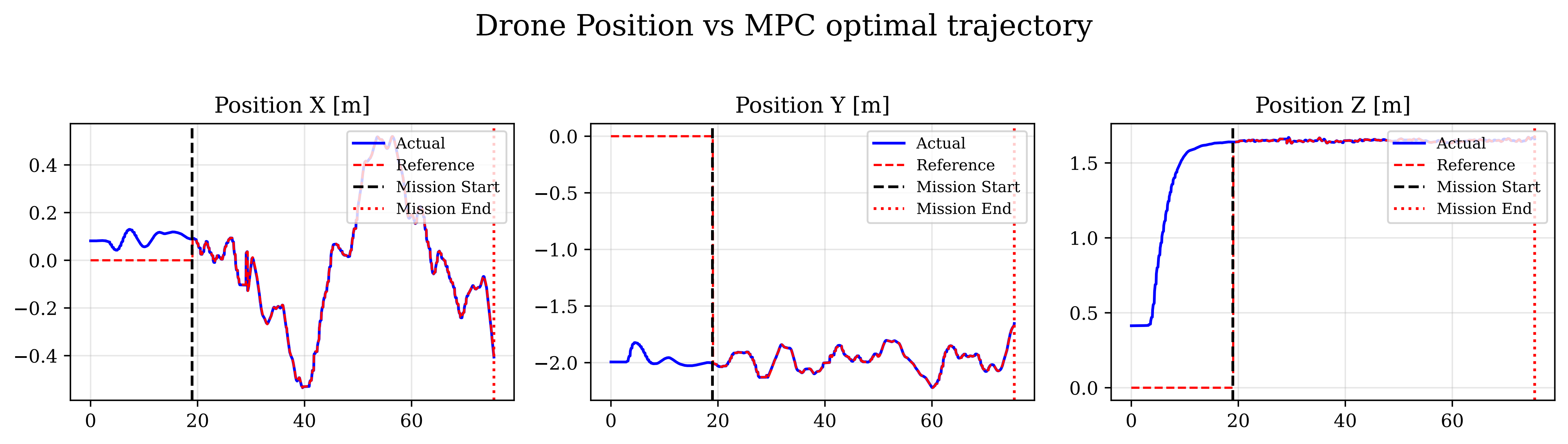
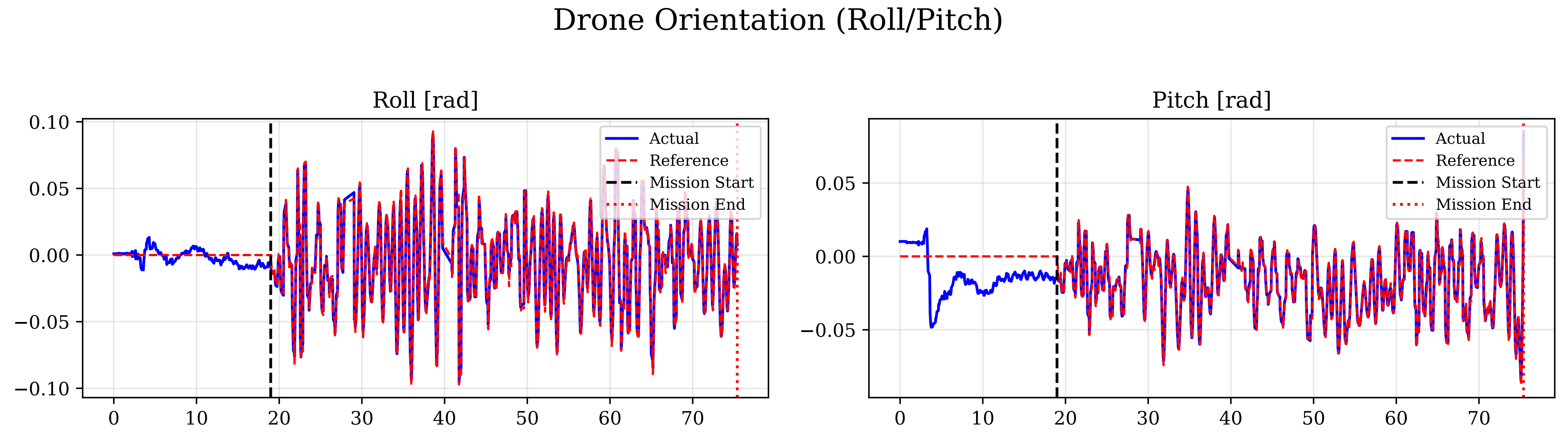
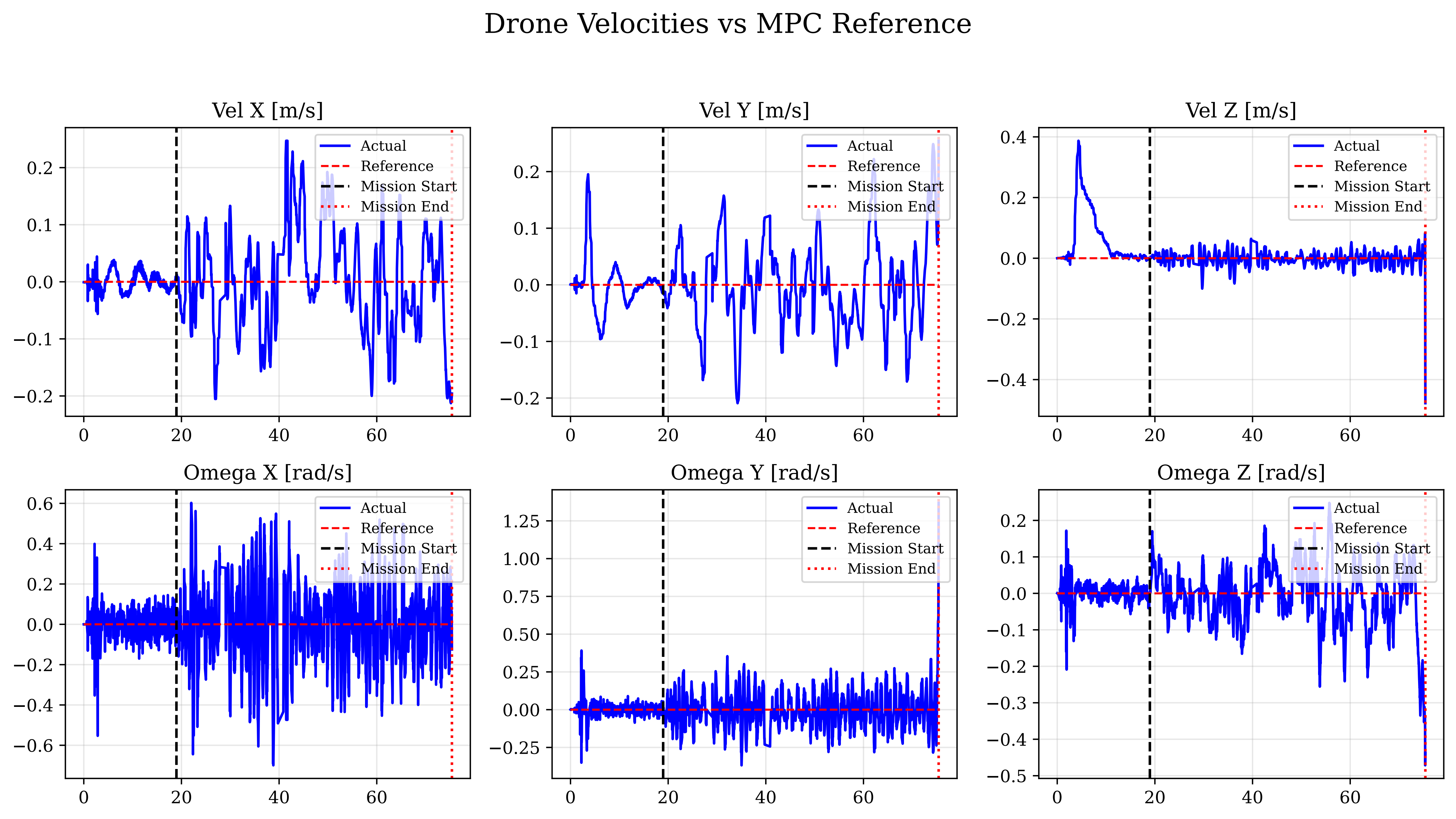
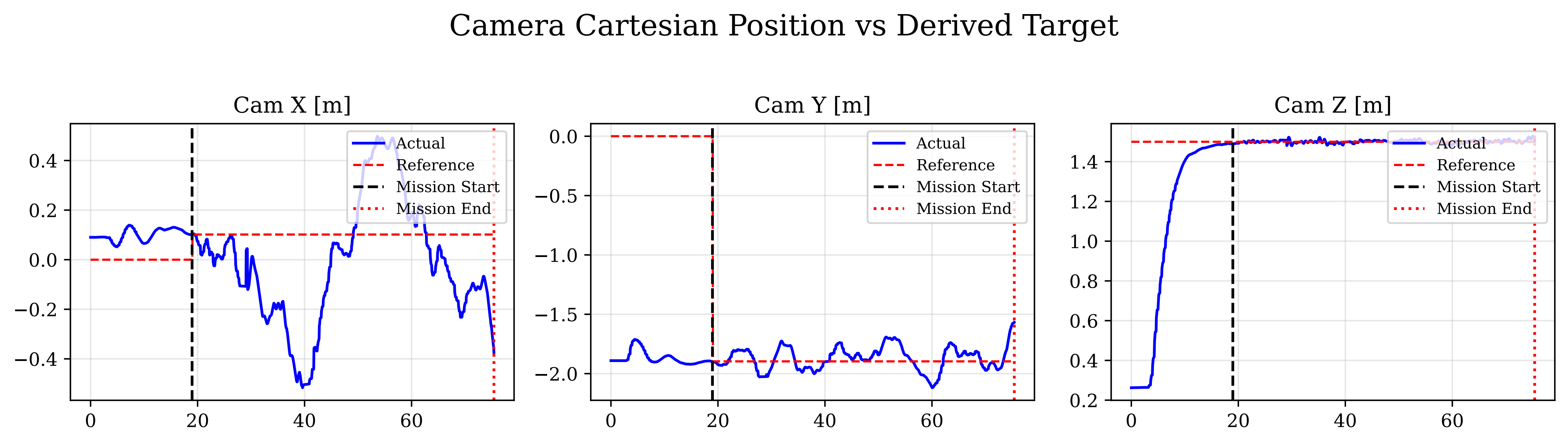


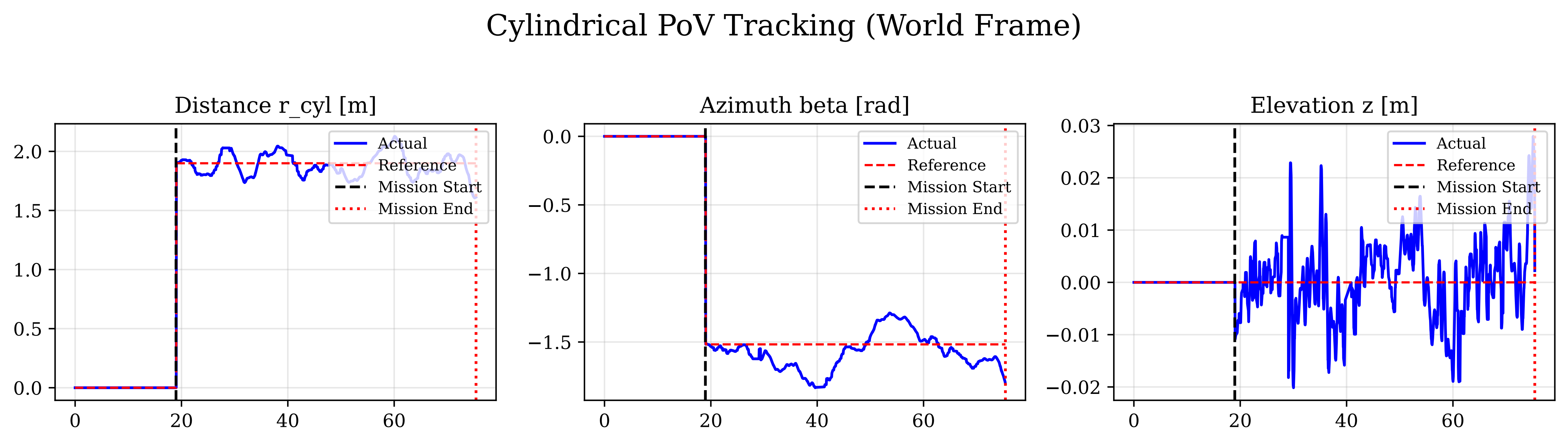
Drone Position vs MPC optimal trajectory
Position X [m]
Position Y [m]
Position Z [m]
Actual
Reference
Mission Start
Mission End
0.4
0.2
0.0
−0.2
−0.4
0.0
−0.5
−1.0
−1.5
−2.0
1.5
1.0
0.5
0.0
0
20
40
60

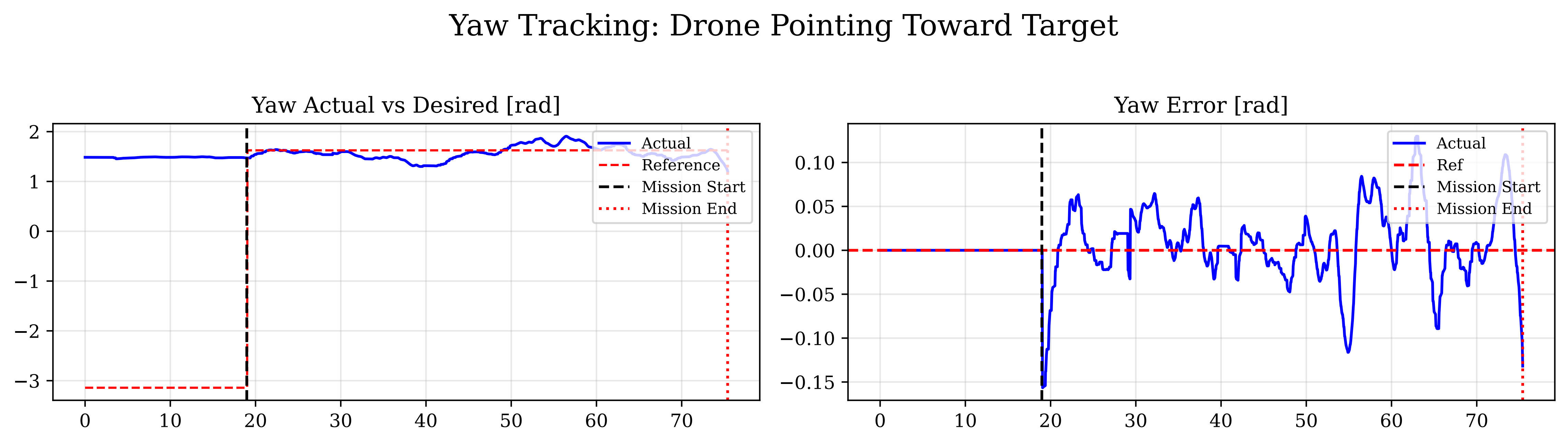

Drone Orientation (Roll/Pitch)
Roll [rad]
Pitch [rad]
Actual
Reference
Mission Start
Mission End
0.10
0.05
0.00
−0.05
−0.10
0
10
20
30
40
50
60
70

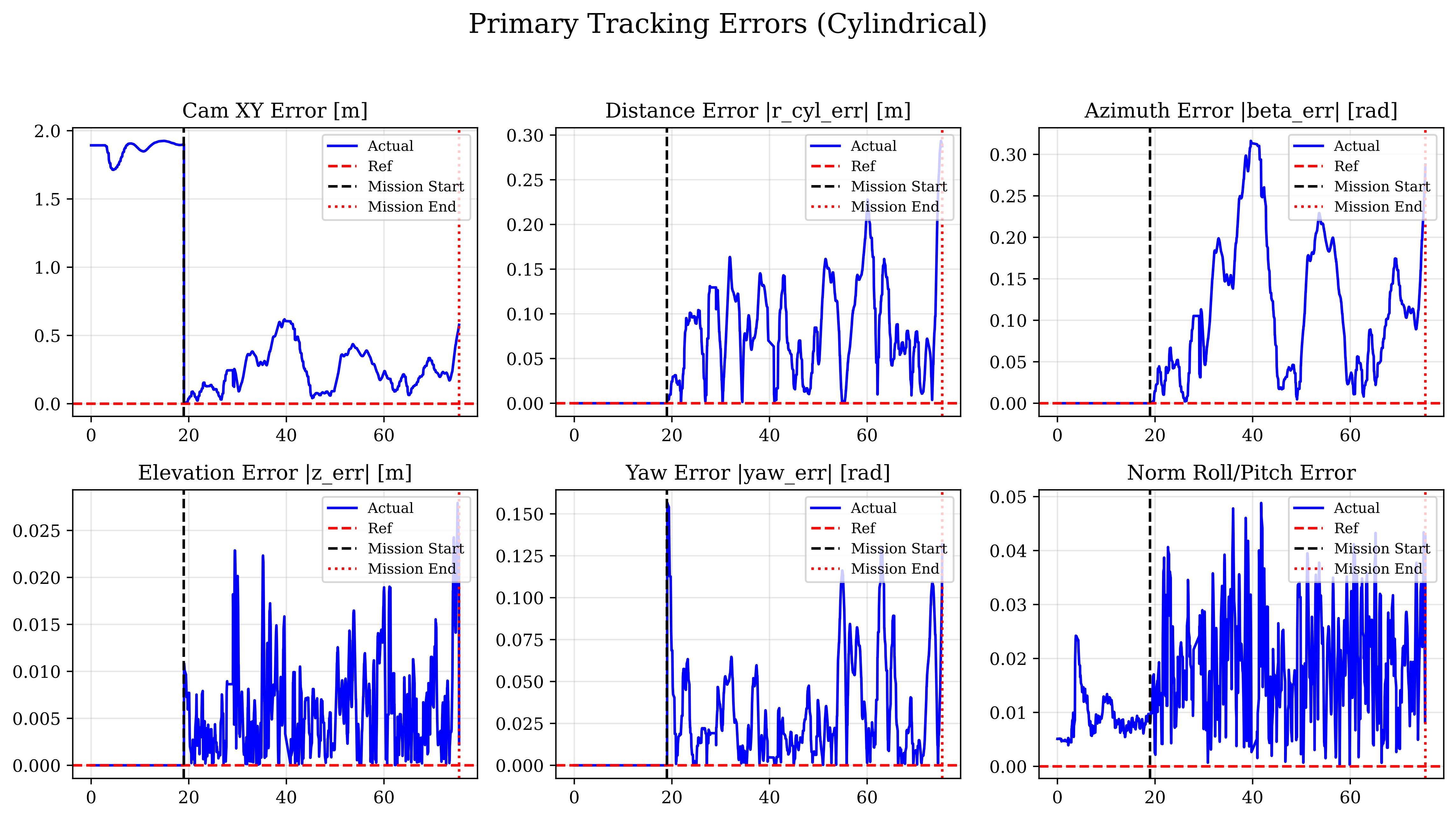

Drone Velocities vs MPC Reference
Vel X [m/s]
Vel Y [m/s]
Vel Z [m/s]
Omega X [rad/s]
Omega Y [rad/s]
Omega Z [rad/s]
Actual
Reference
Mission Start
Mission End



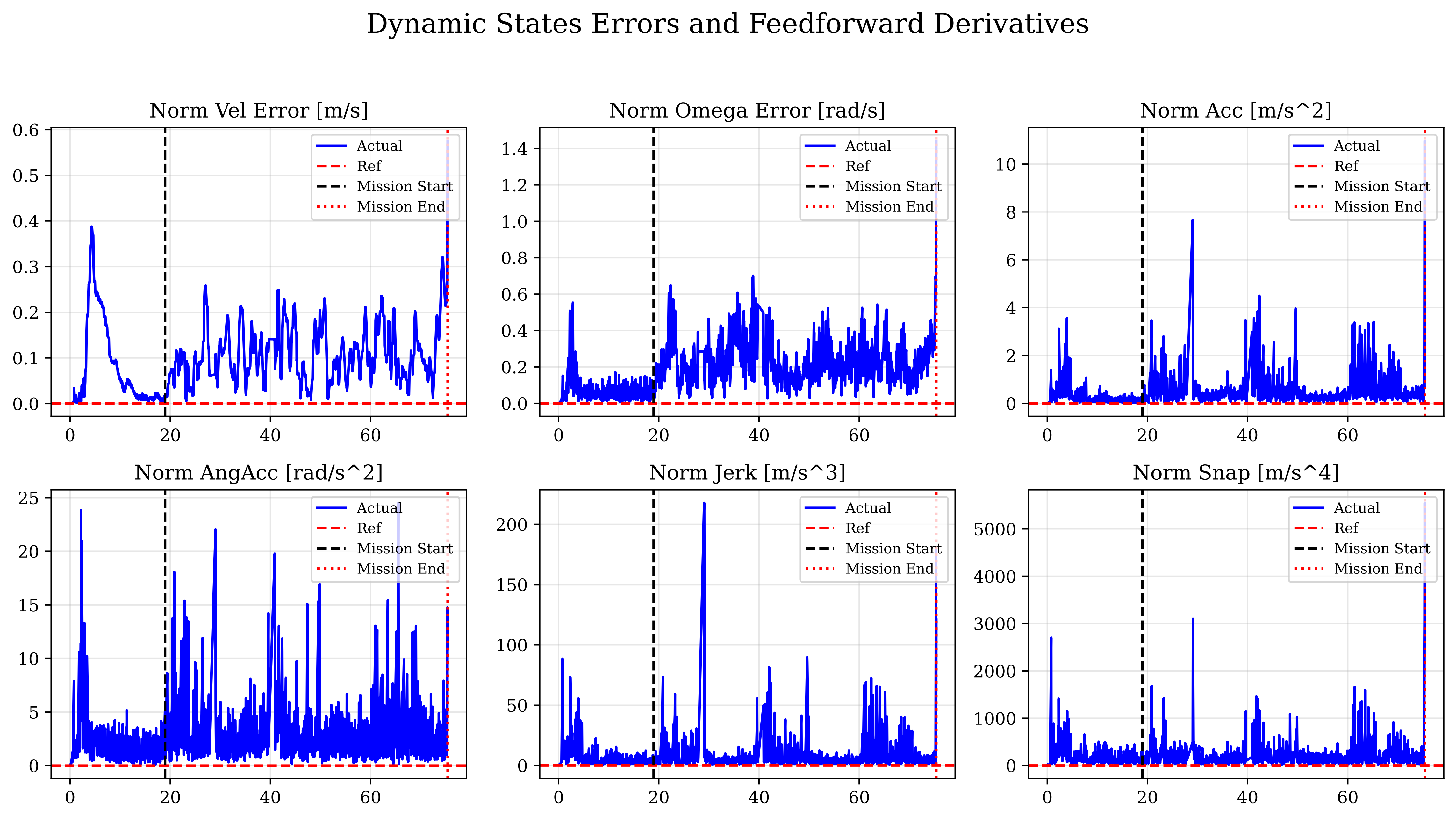
Camera Cartesian Position vs Derived Target
Cam X [m]
Cam Y [m]
Cam Z [m]
Actual
Reference
Mission Start
Mission End
0
20
40
60
0.4
0.2
0.0
−0.2
−0.4
0.0
−0.5
−1.0
−1.5
−2.0
1.4
1.2
1.0
0.8
0.6
0.4
0.2

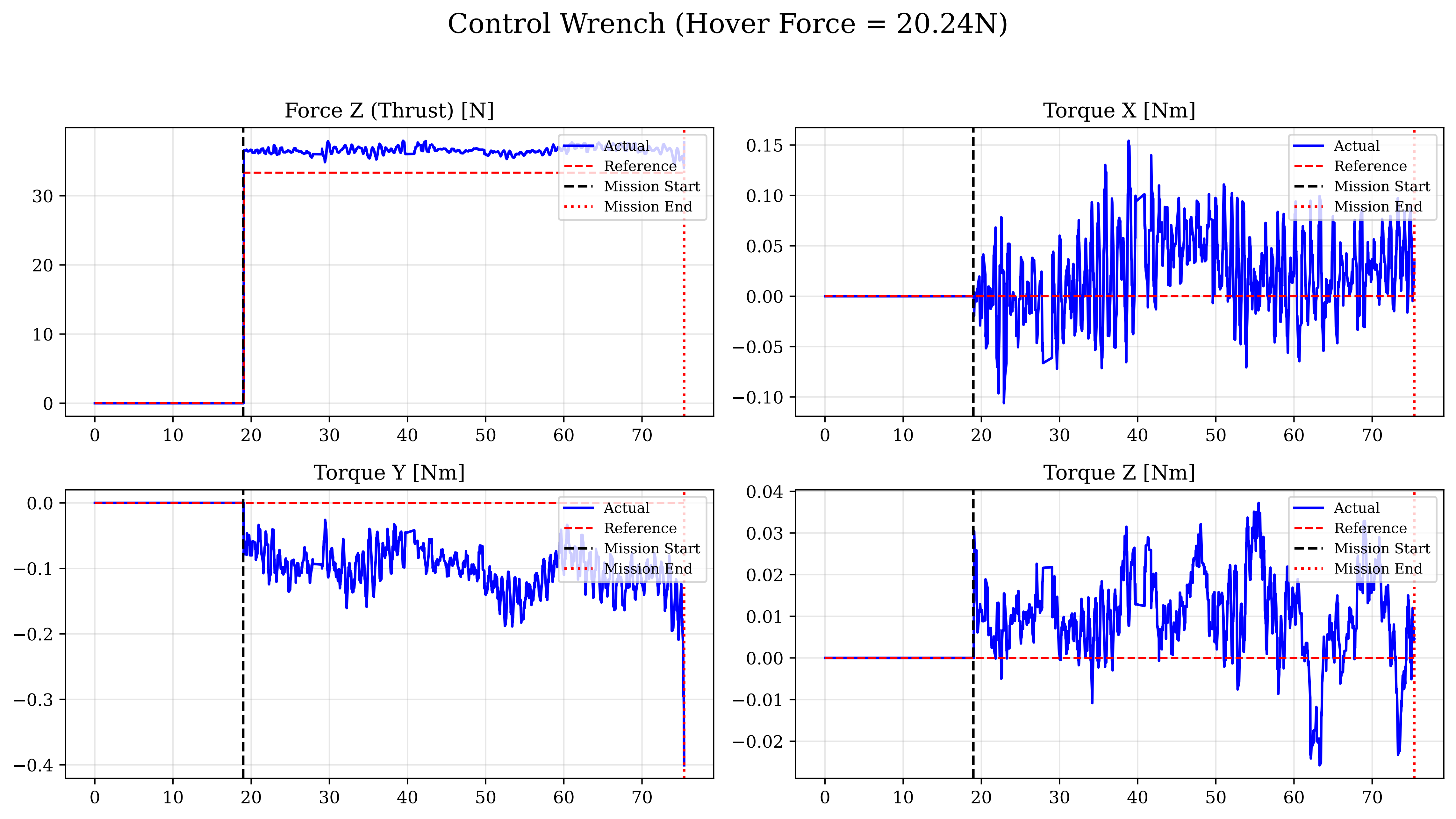

Cylindrical PoV Tracking (World Frame)
Distance r_cyl [m]
Azimuth beta [rad]
Elevation z [m]
Actual
Reference
Mission Start
Mission End
0
20
40
60
0.0
0.5
1.0
1.5
2.0
−0.5
−1.0
−1.5
0.03
0.02
0.01
0.00
−0.01
−0.02


Yaw Tracking: Drone Pointing Toward Target
Yaw Actual vs Desired [rad]
Actual
Reference
Mission Start
Mission End
2
1
0
−1
−2
−3
0
10
20
30
40
50
60
70
Yaw Error [rad]
Actual
Ref
Mission Start
Mission End
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
50
60
70

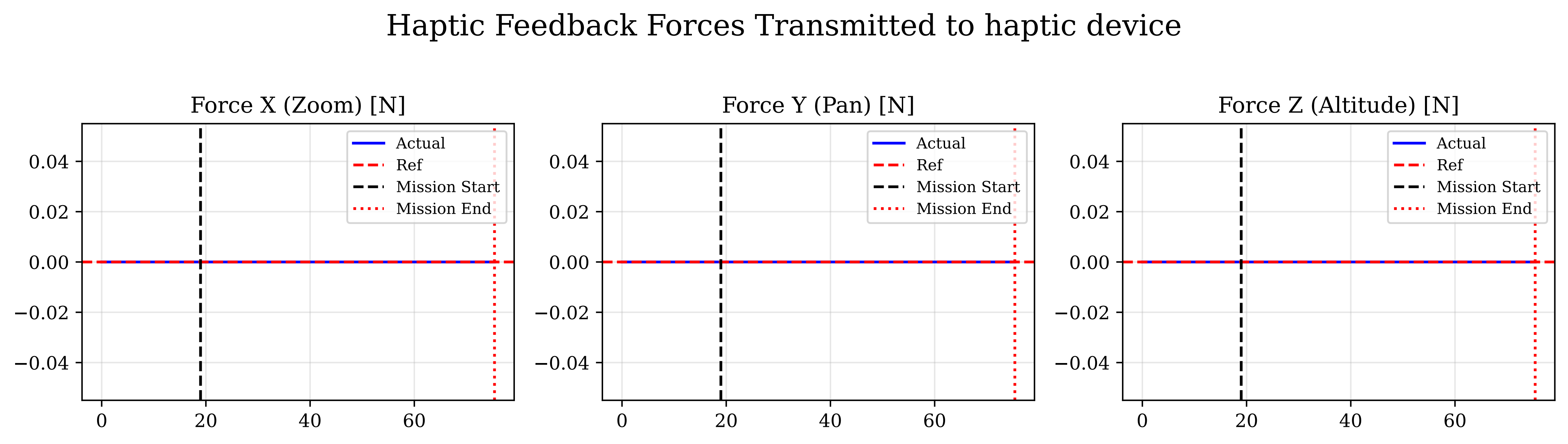


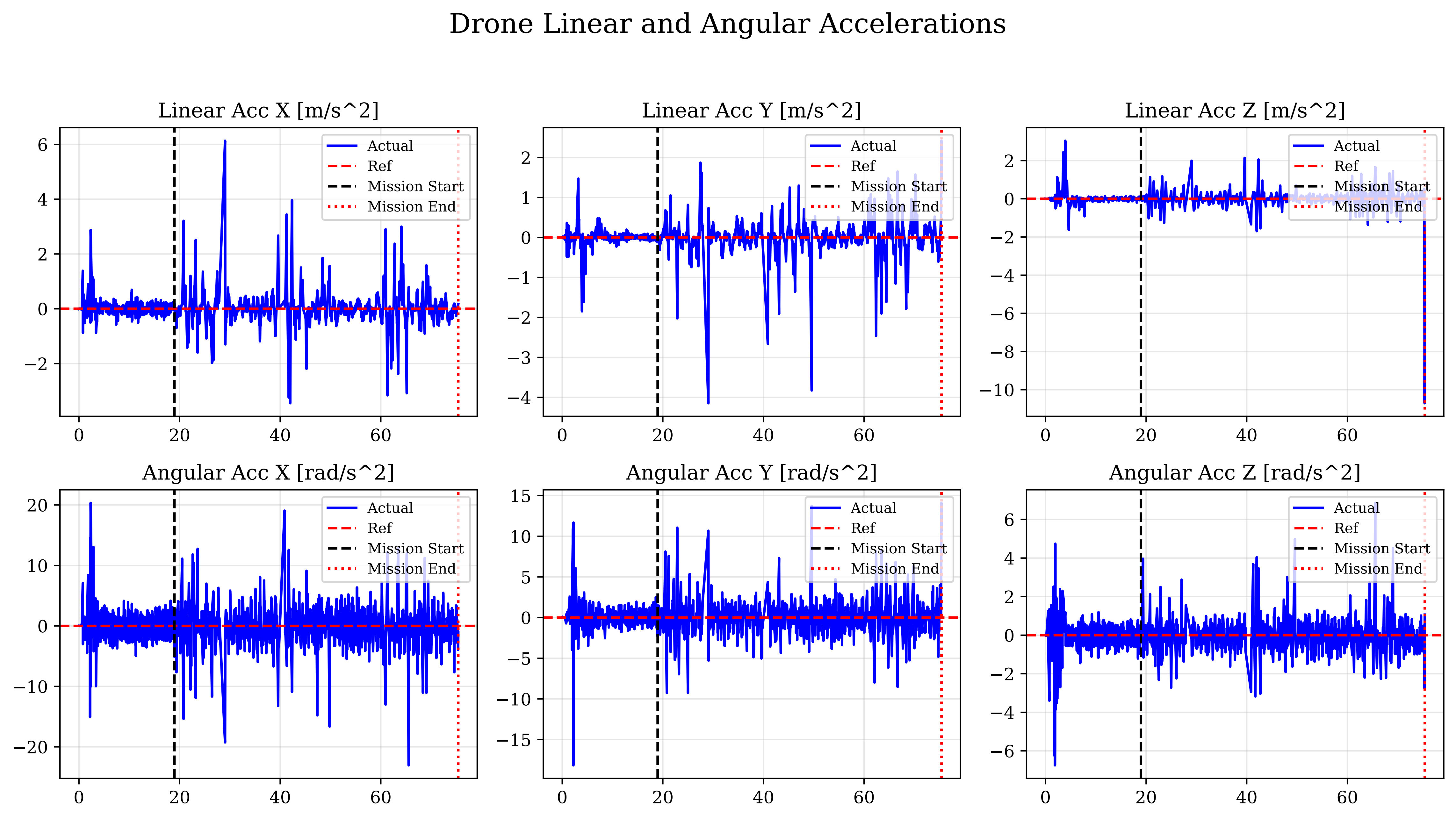
Primary Tracking Errors (Cylindrical)
Cam XY Error [m]
Distance Error |r_cyl_err| [m]
Azimuth Error |beta_err| [rad]
Elevation Error |z_err| [m]
Yaw Error |yaw_err| [rad]
Norm Roll/Pitch Error
Actual
Ref
Mission Start
Mission End

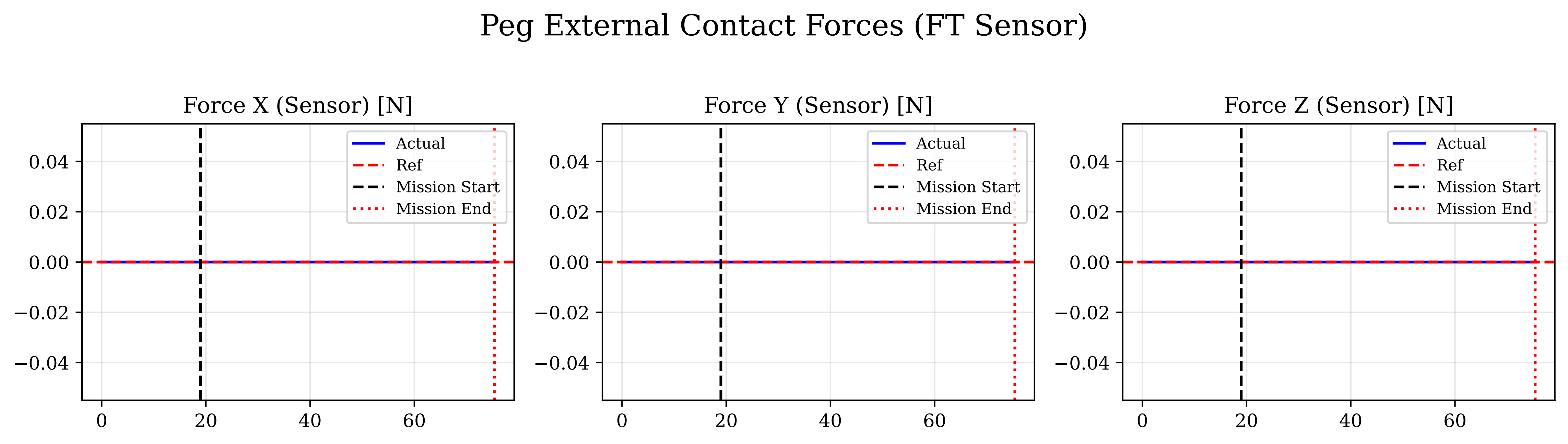

Dynamic States Errors and Feedforward Derivatives
Norm Vel Error [m/s]
Norm Omega Error [rad/s]
Norm Acc [m/s^2]
Norm AngAcc [rad/s^2]
Norm Jerk [m/s^3]
Norm Snap [m/s^4]
Actual
Ref
Mission Start
Mission End



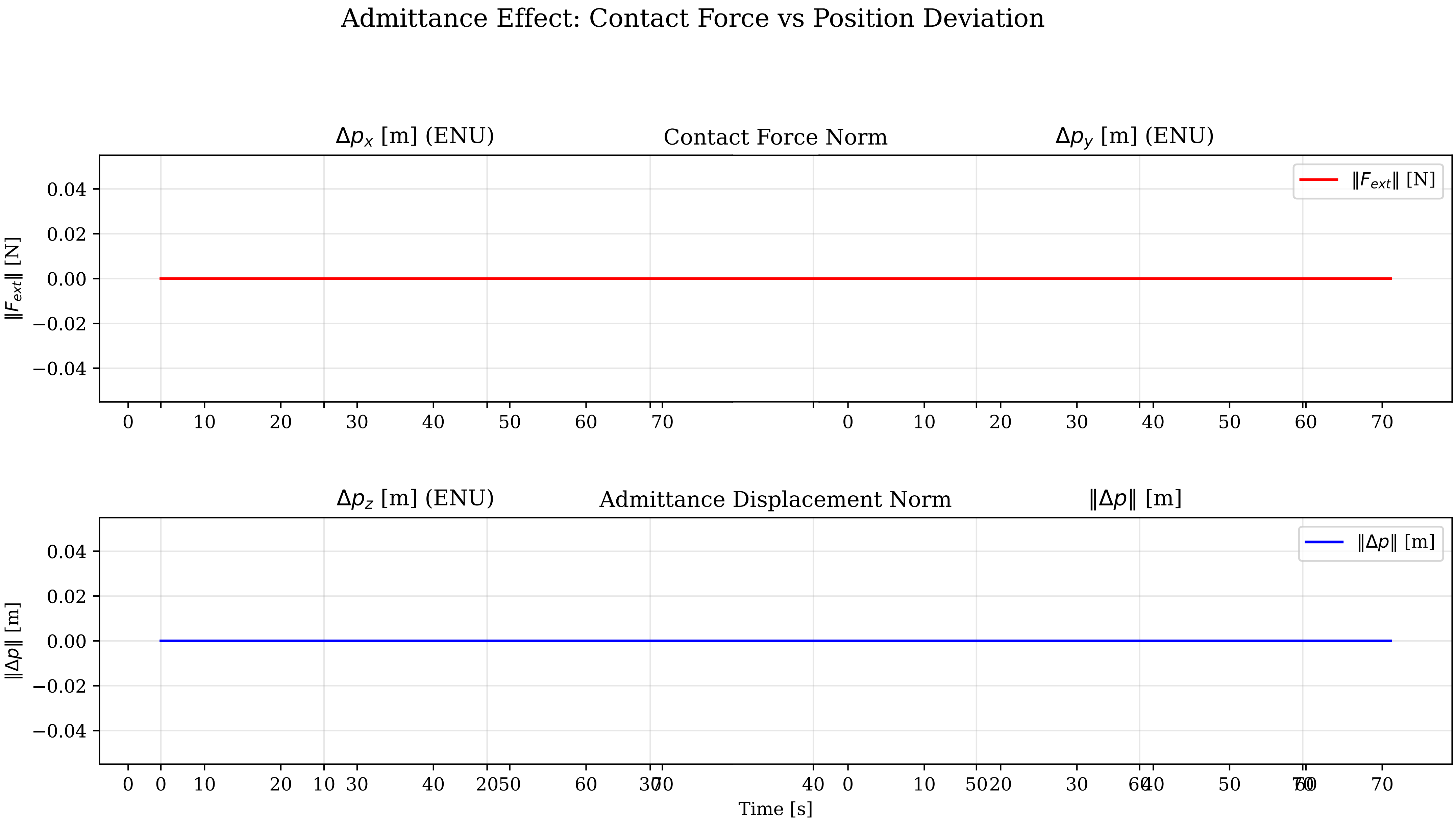
Control Wrench (Hover Force = 20.24N)
Force Z (Thrust) [N]
Torque X [Nm]
Torque Y [Nm]
Torque Z [Nm]
Actual
Reference
Mission Start
Mission End

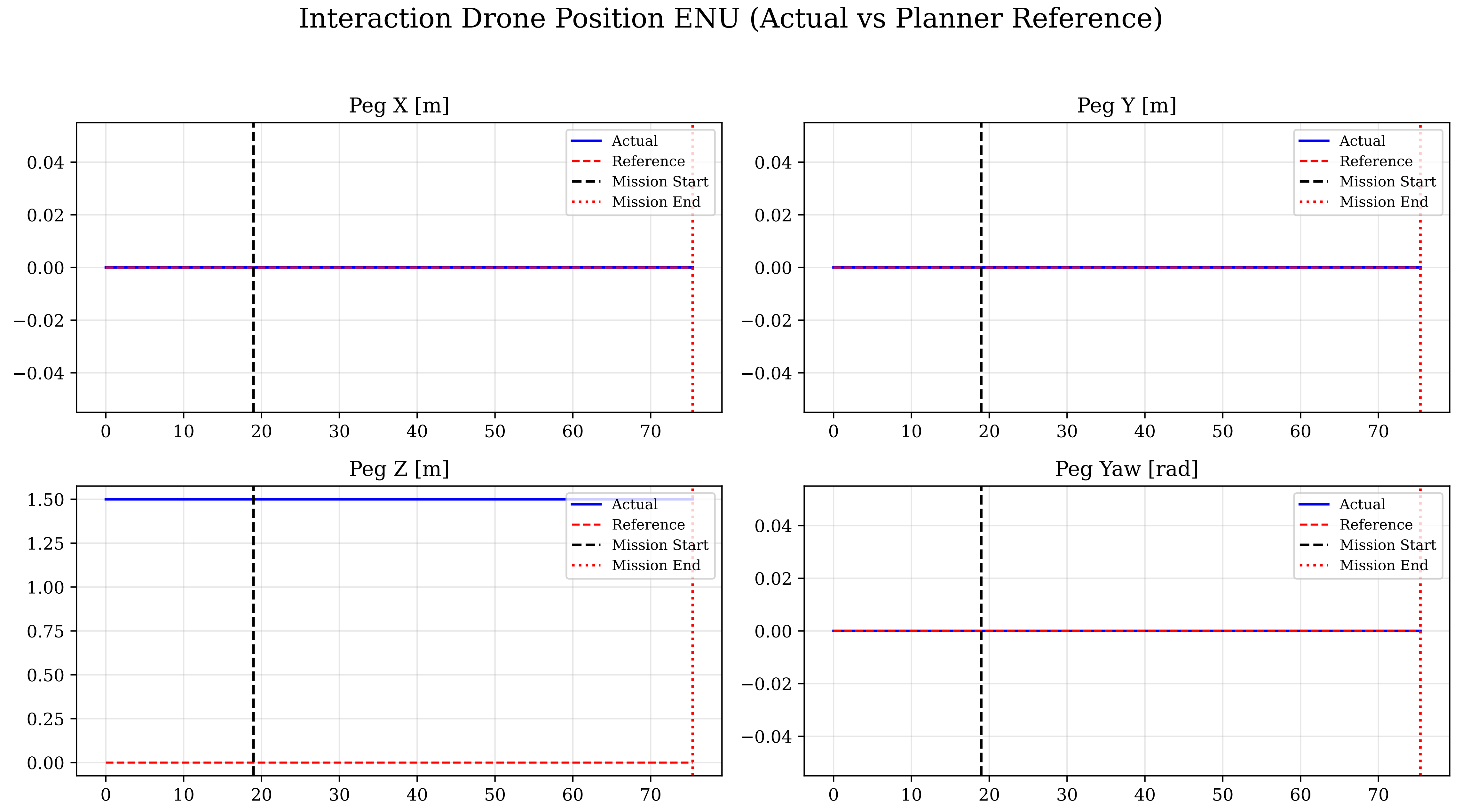

Haptic Feedback Forces Transmitted to haptic device
Force X (Zoom) [N]
Force Y (Pan) [N]
Force Z (Altitude) [N]
Actual
Ref
Mission Start
Mission End
0.04
0.02
0.00
−0.02
−0.04
0
20
40
60


Drone Linear and Angular Accelerations
Linear Acc X [m/s^2]
Linear Acc Y [m/s^2]
Linear Acc Z [m/s^2]
Angular Acc X [rad/s^2]
Angular Acc Y [rad/s^2]
Angular Acc Z [rad/s^2]
Actual
Ref
Mission Start
Mission End

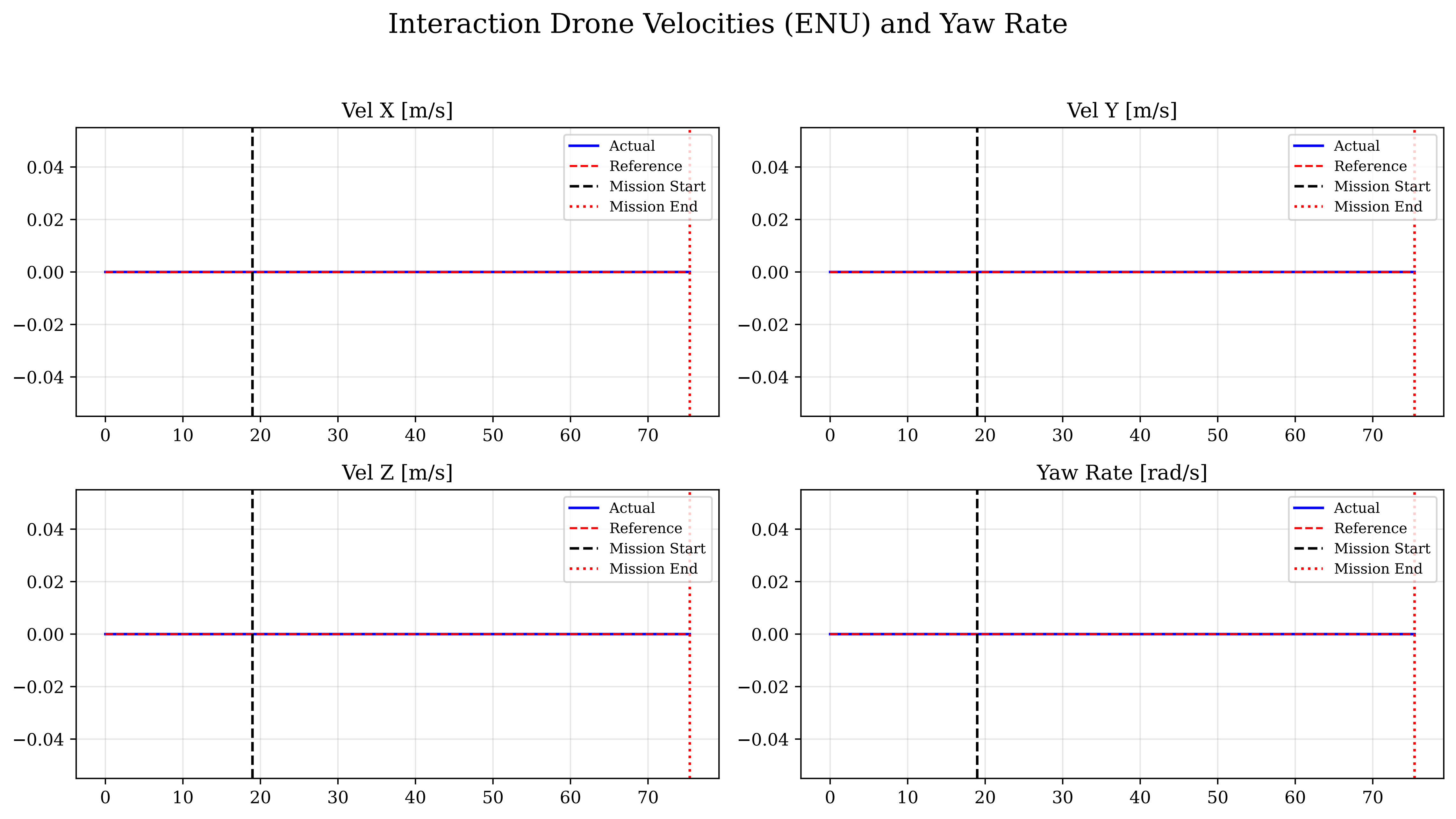


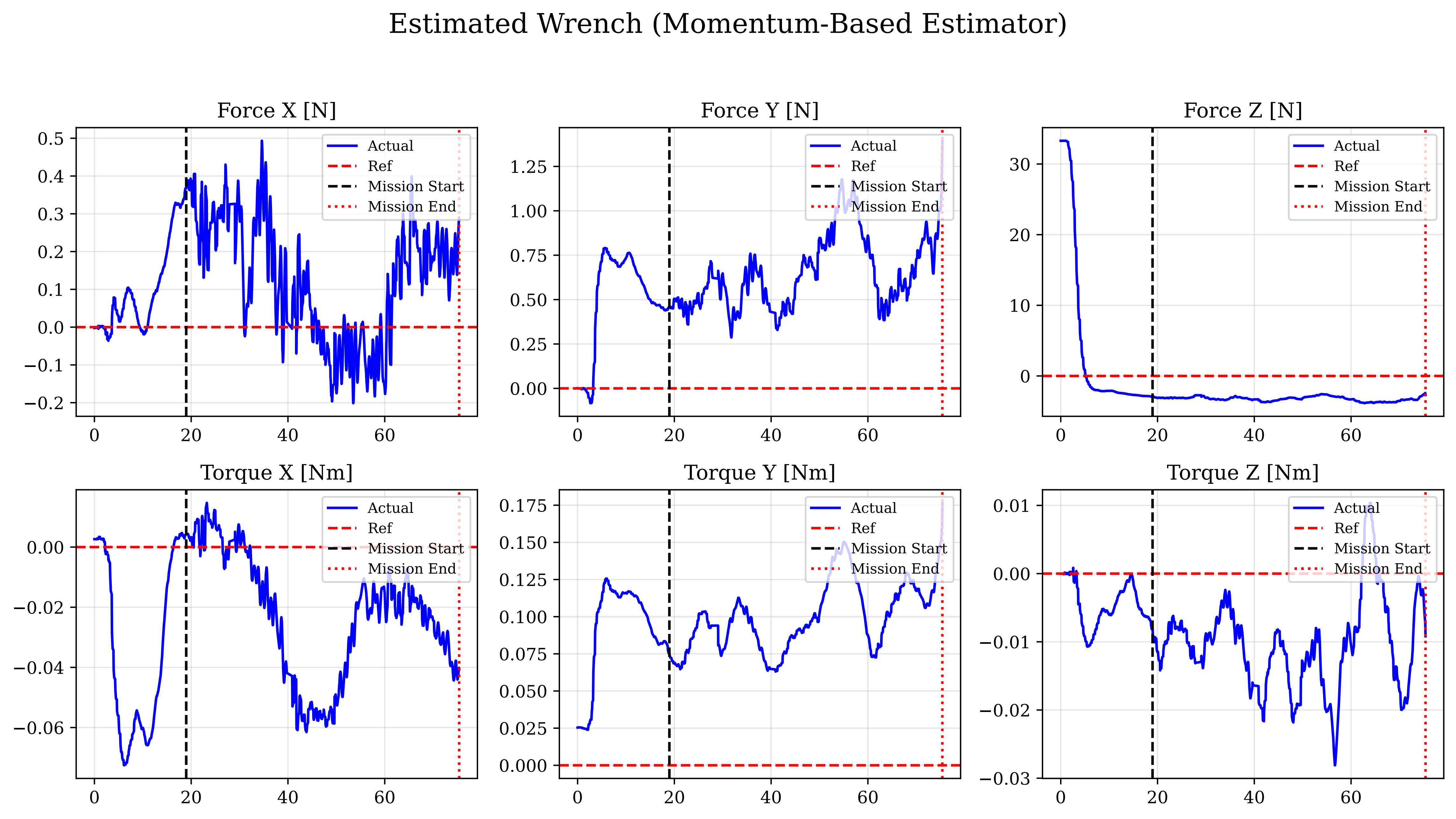
Peg External Contact Forces (FT Sensor)
Force X (Sensor) [N]
Force Y (Sensor) [N]
Force Z (Sensor) [N]
Actual
Ref
Mission Start
Mission End
0.04
0.02
0.00
−0.02
−0.04
0
20
40
60


Admittance Effect: Contact Force vs Position Deviation
$\Delta p_x$ [m] (ENU)
Contact Force Norm
$\Delta p_y$ [m] (ENU)
$\|F_{ext}\|$ [N]
$\|F_{ext}\|$ [N]
$\Delta p_z$ [m] (ENU)
Admittance Displacement Norm
$\|\Delta p\|$ [m]
$\|\Delta p\|$ [m]
$\|\Delta p\|$ [m]
Time [s]

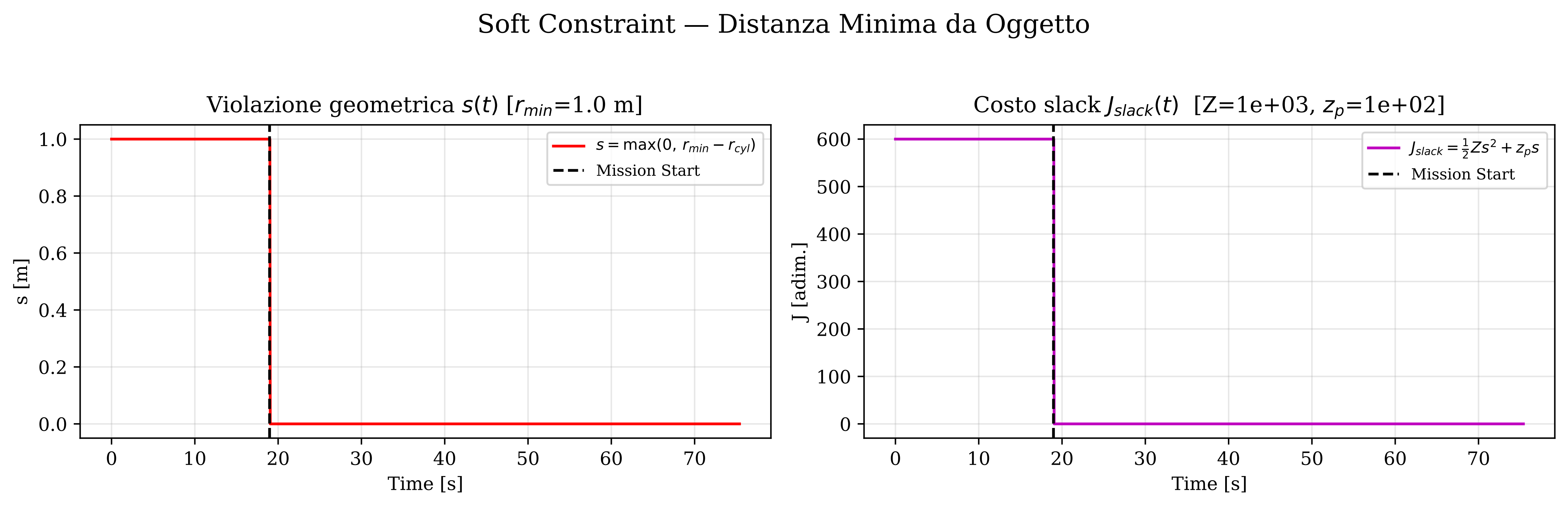


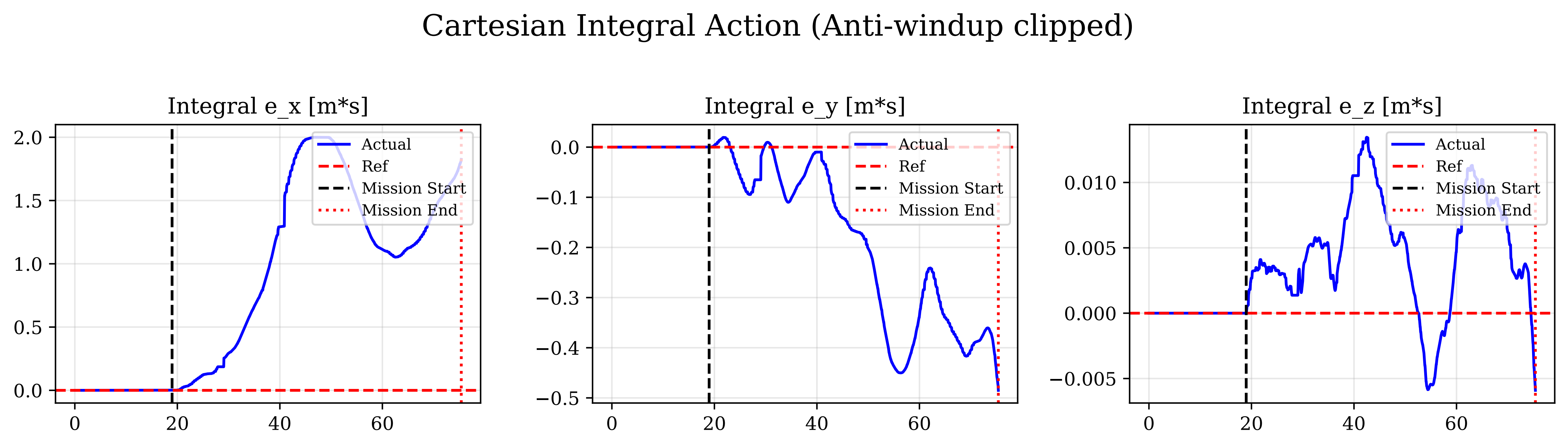
Interaction Drone Position ENU (Actual vs Planner Reference)
Peg X [m]
Peg Y [m]
Peg Z [m]
Peg Yaw [rad]
Actual
Reference
Mission Start
Mission End
0.04
0.02
0.00
−0.02
−0.04
1.50
1.25
1.00
0.75
0.50
0.25
0
10
20
30
40
50
60
70


Interaction Drone Velocities (ENU) and Yaw Rate
Vel X [m/s]
Vel Y [m/s]
Vel Z [m/s]
Yaw Rate [rad/s]
Actual
Reference
Mission Start
Mission End
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40
50
60
70



Estimated Wrench (Momentum-Based Estimator)
Force X [N]
Force Y [N]
Force Z [N]
Torque X [Nm]
Torque Y [Nm]
Torque Z [Nm]
Actual
Ref
Mission Start
Mission End


Soft Constraint — Distanza Minima da Oggetto
Violazione geometrica $s(t)$ [$r_{min}$=1.0 m]
$s = \max(0,\ r_{min} - r_{cyl})$
Mission Start
s [m]
Time [s]
Costo slack $J_{slack}(t)$ [Z=1e+03, $z_p$=1e+02]
$J_{slack} = \frac{1}{2}Zs^2 + z_p s$
Mission Start
J [adim.]
Time [s]


Cartesian Integral Action (Anti-windup clipped)
Integral e_x [m*s]
Integral e_y [m*s]
Integral e_z [m*s]
Actual
Ref
Mission Start
Mission End
0
20
40
60
0.0
0.5
1.0
1.5
2.0
−0.5
−0.4
−0.3
−0.2
−0.1
−0.005
0.000
0.005
0.010

Admittance Displacement in Sensor Frame